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Inventor(s)

TAHIR; Muhammad

METHOD AND SYSTEM FOR CONSTRUCTING CFs/TiO₂ NANOTEXTURE FROM RECYCLED CARBON FIBER-REINFORCED POLYMERS (CFRPs) FOR PHOTOCATALYTIC HYDROGEN PRODUCTION

Abstract

There is disclosed a method of developing semiconductor photocatalysts by recycling Carbon Fiber-Reinforced Polymers (CFRP) waste, the method comprising separating or removing a polymer layer of carbon-fibre waste via thermal decomposition; and coupling the resulting carbon-fibres as a cocatalyst with semiconductor materials; for photocatalytic water splitting results in producing hydrogen (H₂). The semiconductor materials such as titanium dioxide (TiO₂), to be used as composite materials, and coupling the carbon-fibres as a cocatalyst with semiconductor materials is done via facile hydrothermal methods and ultrasonic/physical mixing approaches (CFs/TiO₂-A). Further disclosed is a carbon fiber (CF)/TiO₂ composite comprising well-distributed and uniformly sized TiO₂ nanoparticles, wherein the TiO₂ particles are uniformly attached to the CF surface, wherein the CF is synthesized by being separated from a polymer layer of CFRP waste.

Inventors: TAHIR; Muhammad (Al Ain, AE)

Applicant: United Arab Emirates University (Al Ain, AE)

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Background/Summary

FIELD OF THE INVENTION

[0001] The present invention relates to the field of developing efficient and low-cost semiconductor photocatalysts, and more particularly to a system and method of recycling Carbon Fiber-Reinforced Polymers (CFRPs) waste to produce efficient carbon-fibres (CFs) as a cocatalyst to construct CFs supported titanium dioxide (TiO₂) nanotextures.

BACKGROUND OF THE INVENTION

[0002] Background description includes information that may be useful in understanding the present invention. It is not an admission that any of the information provided herein is prior art or relevant to the presently claimed invention, or that any publication specifically or implicitly referenced is prior art.

[0003] Hydrogen stands out as a versatile and environmentally friendly energy source with the potential to mitigate the reliance on fossil fuels and curb greenhouse gas emissions. Particularly appealing is the prospect of harnessing solar energy for photocatalytic hydrogen (H₂) production, offering a sustainable alternative devoid of carbon dioxide or other harmful pollutants. However, realizing the full potential of this technology necessitates the development of efficient, cost-effective, and stable photocatalysts. A promising photocatalyst should have high efficiency for absorbing light radiation and promoting the desired chemical reaction with good stability.

[0004] Among the many semiconductors explored for H₂ production, Titanium dioxide (TiO₂) emerges as a promising candidate due to its abundance, affordability, chemical stability, non-toxicity, and higher oxidation potential. However, it has some limitations, such as a wide band gap and a high electron-hole recombination rate, which affect its photocatalytic activity.

[0005] Various strategies have been explored to enhance the photoactivity of semiconductors, including doping with metals, coupling with cocatalysts, and hybridization with other semiconductors. Notable investigations include Au/TiO₂, g-C₃N₄-reinforced CdS nanosphere-decorated TiO₂, NiS/Pt-loaded TiO₂, Pd modified ZnIn₂S₄/P25, Cu₃P/TiO₂, 0D Co₃O₄/1D TiO₂, and Ti₃C₂

MXene@TiO₂/CuInS₂ composites, aimed at augmenting TiO₂'s photocatalytic activity for hydrogen production.

[0006] Carbon materials, particularly carbon fibers (CFs), have gained significant attention across industries like aerospace, automotive, and sports due to their remarkable mechanical, thermal, and chemical properties. However, excessive use of Carbon fiber-reinforced polymers (CFRPs) has created environmental concerns due to the generation of scraps and end-of-life components and an economic concern to recycle CFRP waste.

[0007] However, despite advancements in this field, there exists a gap in the literature regarding the utilization of CFRP waste for constructing CFs/TiO₂ composites with different nanotextures for stimulating photocatalytic hydrogen production.

[0008] The present disclosure seeks to overcome one or more of these problems. More specifically, but not exclusively, the present disclosure seeks to provide a solution to overcome the disadvantages and shortcomings of developing composites that may improve semiconductor performance in solar energy-related applications like photocatalytic green hydrogen production.

SUMMARY OF THE INVENTION

[0009] Therefore it is an object of the present invention to propose the utilization of carbon fiber-

reinforced polymer (CFRP) waste-derived carbon fibers (CFs) coupled with TiO₂ for photocatalytic hydrogen production. The CFs are effectively used to enhance visible light absorption and to prevent photoinduced charge carrier recombination during photocatalytic hydrogen production.

[0010] In an embodiment of the invention, nanocomposites are synthesized via physical mixing (CFs/TiO₂-A) and sol-gel methods (CFs/TiO₂-S) to explore the role of interface interaction and morphology.

[0011] There is disclosed a method of developing semiconductor photocatalysts by recycling Carbon Fiber-Reinforced Polymers (CFRP) waste, the method comprising the steps of separating or removing a polymer layer of carbon-fibre waste via thermal decomposition; and coupling the resulting carbon-fibres as a cocatalyst with semiconductor materials: wherein combining sewage sludge derived from the CFRP waste with semi-conductors for photocatalytic water splitting results in producing/generating hydrogen (H₂).

[0012] In an embodiment of the invention, carbon-fibres (CFs) are synthesized by being separated from the polymer layer via thermal decomposition of CFRPs, at low temperatures (400 to 700° C.) and heating times (1 to 4 hours), yielding highly conductive carbon fibers (CFs).

[0013] In another embodiment of the invention, the CFs are tailored into various sizes and shapes by eliminating the polymer layer.

[0014] In another embodiment of the invention, the CFRP serving as the raw material is subjected to pyrolysis, the method comprising evenly cutting CFRP sheets into smaller sized rectangular shapes (2 to 6 cm length and 1 to 4 cm width): placed in a ceramic crucible: heating the CFRPs in a tube furnace under controlled nitrogen atmosphere for removing polymeric material; and grinding the pyrolyzed product and pulverizing the ground product into fine powder.

[0015] In another embodiment of the invention, the controlled nitrogen atmosphere comprises a temperature of 500° C. for 2 hours.

[0016] In another embodiment of the invention, pulverizing the ground product into fine powder is done using a planetary ball mill.

[0017] In another embodiment of the invention, coupling the resulting carbon-fibres as a cocatalyst with semiconductor materials is done via facile hydrothermal methods such as sol-gel (CFs/TiO₂-S) and ultrasonic/physical mixing approaches (CFs/TiO₂-A).

[0018] In an embodiment, using the sol-gel approach. TiO₂ is attached over the entire surface of CFs, enabling interface interaction and charge carrier separation, resulting in enhanced H₂ production.

[0019] In an embodiment, using the sol-gel approach results in achieving a core-shell structure of nanoparticles with carbon fibres.

[0020] In an embodiment, the CFRP waste-derived carbon fibers (CFs) is coupled with TiO₂ to construct CFs/TiO₂ nanotextures with efficient interface charge transfer for stimulating photocatalytic hydrogen production.

[0021] As another aspect of the proposed invention, there is disclosed a carbon fiber (CF)/TiO₂ composite comprising well-distributed and uniformly sized TiO₂ nanoparticles, wherein the TiO₂ particles are uniformly attached to the CF surface, to be used as a composite material for photocatalytic water splitting to produce green hydrogen and CO₂ emission reduction, wherein the carbon-fibre (CF) is synthesized by being separated from a polymer layer of Carbon Fiber-Reinforced Polymers (CFRP) waste.

[0022] In another embodiment, the TiO₂ nanoparticles are spherical shaped, without agglomeration formations.

[0023] In an embodiment, the CF/TiO₂ composite is synthesized using a facile hydrothermal method such as the sol-gel method.

[0024] In an embodiment, the CF/TiO₂ composite is synthesized using physical mixing approaches.

[0025] In an embodiment, the morphology of the CFs/TiO.sub.2 is investigated using methods such as transmission electron microscopy (TEM), X-ray powder diffraction (XRD), Raman spectroscopy and scanning electron microscopy (SEM).

[0026] In another embodiment, the CF/TiO.sub.2 composite promotes charge separation and visible light absorption during the photocatalysis process.

[0027] In an embodiment, glycerol is used as a sacrificial reagent to improve hydrogen (H.sub.2) yield.

[0028] In an embodiment, an apparent quantum yield (AQY) of 26.3% for H.sub.2 production is obtained with 150 mg of catalyst loading under low-intensity light irradiation.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0029] The subject matter that is regarded as the invention is particularly pointed out and distinctly claimed in the claims at the conclusion of the specification. The foregoing and other aspects, features, and advantages of the invention are apparent from the following detailed description taken in conjunction with the accompanying drawings in which:

[0030] FIG. 1 shows the schematic illustration for the preparation of CFs from CFRPs through thermal decomposition, in accordance with the present invention.

[0031] FIG. 2 is a schematic illustration for the physical mixing method for CFs/TiO.sub.2 composites, in accordance with an embodiment of the present invention.

[0032] FIG. 3 shows the sol-gel method for the synthesis of CFs/TiO.sub.2—S composite, in accordance with another embodiment of the present invention.

[0033] FIG. 4 shows XRD patterns of CF, TiO.sub.2-A, TiO.sub.2—S, CF/TiO.sub.2-A, and CF/TiO.sub.2—S samples.

[0034] FIG. 5A shows Raman analysis of CFs according to an embodiment of the present disclosure.

[0035] FIG. 5B shows Raman patterns of TiO.sub.2-A, TiO.sub.2—S, CF/TiO.sub.2-A, and CF/TiO.sub.2—S samples.

[0036] FIG. 6 shows IR patterns of CF, TiO.sub.2-A, TiO.sub.2—S, CF/TiO.sub.2-A, and CF/TiO.sub.2—S samples, according to an embodiment of the present disclosure.

[0037] FIG. 7A, FIG. 7B, FIG. 7C-7D, FIG. 7E-7F, and FIG. 7G-7H show SEM images of TiO2-A, TiO2-S, CFs, 3CFs/TiO2-A, and 3CFs/TiO2-S respectively, according to the invention.

[0038] FIG. 8A and FIG. 8B show XPS spectra of Ti 2p and C 1s respectively.

[0039] FIG. 8C shows wide spectra for TiO.sub.2 VB calculation, in accordance with an embodiment of the present invention.

[0040] FIG. 9 shows UV-vis diffuse reflectance spectra of TiO.sub.2, CFs, and their composites.

[0041] FIG. 10 is PL analysis of CF, TiO.sub.2-A, TiO.sub.2—S, and CF/TiO.sub.2-A and CF/TiO.sub.2—S composite samples, according to an embodiment of the present disclosure.

[0042] FIG. 11 shows N.sub.2 adsorption-desorption isotherms of TiO.sub.2—S and CF/TiO.sub.2—S composite.

[0043] FIG. 12A presents the effect of CF-loading on the performance of TiO.sub.2-A for photocatalytic H.sub.2 evolution in a 5 vol % methanol solution and 100 mg catalyst loading, in accordance with an embodiment of the present invention.

[0044] FIG. 12B shows performance comparison of 3CF/TiO.sub.2-A and 3CF/TiO.sub.2—S synthesized through sol-gel method for photocatalytic H.sub.2 evolution with 5 vol % methanol and 100 mg catalyst loading.

[0045] FIG. 12C is a schematic illustration for the generation of H.sub.2 for the catalysts synthesized using physical mixing, according to an embodiment of the present disclosure.

[0046] FIG. 12D is schematic illustration of charge transfer over CFs/TiO₂-S produced using sol-gel approach, according to another embodiment of the present invention.

[0047] FIG. 13 shows the effect of sacrificial reagents on the performance of the 3CFs/TiO₂-S composite catalyst.

[0048] FIG. 14 shows the effect of CFs loading on the performance of 3CFs/TiO₂-S for photocatalytic H₂ evolution using a glycerol sacrificial reagent.

[0049] FIG. 15A presents photostability of 3CFs/TiO₂-S with 5% glycerol for continuous H₂ evolution for 5 consecutive cycles with 150 mg catalyst loading, according to an embodiment of the present invention.

[0050] FIG. 15B presents photostability of 3CFs/TiO₂-S with 5% methanol for continuous H₂ evolution for consecutive 5 cycles with 150 mg catalyst loading, according to another embodiment of the present invention.

[0051] FIG. 16 shows XRD analysis of the fresh and the spent catalyst.

[0052] FIG. 17 shows FTIR analysis of 3% CFs/TiO₂-S composite after the reaction of methanol and glycerol.

[0053] FIG. 18 shows EDX spectra of 3CFs/TiO₂-S.

[0054] FIG. 19 illustrates proposed mechanism for photocatalytic H₂ evolution over a 3CFs/TiO₂-S composite photocatalyst, according to the present invention.

DETAILED DESCRIPTION OF THE INVENTION

[0055] The aspects of the proposed system and method for recycling CFRPs to produce CFs as cocatalyst to be coupled with TiO₂ for efficient photocatalytic H₂ production according to the present invention will be described in conjunction with FIGS. 1-19. In the Detailed Description, reference is made to the accompanying figures, which form a part hereof, and which is shown by way of illustration specific embodiments by which the invention may be practiced. It is to be understood that other embodiments may be utilized and logical changes may be made without departing from the scope of the present invention. The following detailed description, therefore, is not to be taken in a limiting sense, and the scope of the present invention is defined by the appended claims.

[0056] Hydrogen is a versatile and clean energy source. The utilization of solar energy for generating hydrogen (H₂) through photocatalytic water splitting has emerged as a highly promising technology, that offers a sustainable solution for renewable fuel production with reduced carbon dioxide (CO₂) emissions. This technology, however, requires the development of efficient, low-cost, and stable photocatalysts. The proposed invention explores solid waste carbon materials such as carbon fibre-reinforced polymers (CFRPs) to be utilized as an efficient cocatalyst for photocatalytic green H₂ production using solar energy. In this process, novel low-cost nanomaterials of tenable size and shape are synthesized using a template-free route and are characterized for applications. These newly developed composites are found very efficient for producing hydrogen and CO₂ reduction via low-intensity solar energy for renewable fuels.

[0057] The present invention discloses highly efficient and low-cost semiconductor photocatalysts, and more particularly to CFs-based cocatalysts produced from recycled carbon fiber-reinforced polymer (CFRP) waste for solar energy applications, particularly in photocatalytic hydrogen production. These waste-derived CFs can be combined as a cocatalyst with different semiconductors to be used as composite materials. In an embodiment, carbon fiber-reinforced polymer (CFRP) waste-derived carbon fibers (CFs) is coupled with TiO₂ to construct CFs/TiO₂ nanotextures with efficient interface charge transfer for stimulating photocatalytic hydrogen production. Coupling with CFs is achieved using sol-gel and ultrasonic approach. The newly developed composites exhibit a notably enhanced photoactivity and stability, in comparison to the semiconductors.

[0058] The invention also relates to a method of recycling CFRP waste to produce CFs as a cocatalyst to construct CFs/TiO₂ nanotextures for photo-catalytic hydrogen production and

other solar energy-related applications. The method proposes to produce carbon fibres using solid waste materials comprising CFRPs using a facile hydrothermal method. The solid CFRP waste can be converted to pure carbon fibers (CFs) which can be used as highly conductive materials in semiconducting applications. In an embodiment of the invention, the Carbon fibres (CFs) are synthesized/obtained through a low-cost and easy method, wherein they are separated from the polymer layer via thermal decomposition of CFRPs. The process of thermal decomposition at low temperatures yields highly conductive carbon fibers, which can be tailored into various sizes and shapes by eliminating the polymer layer.

[0059] When combined with semiconductors, CFs augment the photoactivity of semiconductors by improving visible light absorption and preventing the recombination of photoinduced charge carriers. This is because CFs are a metal-free source to trap and transport electrons within the semiconductor. Carbon fibers coated with TiO₂ thus, enhances hydrogen production in photocatalytic water splitting due to the narrowed band gap energy and efficient separation of photogenerated electron-hole pairs. In addition, they can be used as cocatalysts because of their high surface area, great electrical conductivity, affordability, and reusability. The CF-supported composites are also strong and durable with improved photostability.

[0060] The role of interface interaction and morphology is investigated by synthesizing nanocomposites through physical mixing (CFs/TiO₂) and sol-gel methods (CFs/TiO₂—S). The optimized 3CFs/TiO₂ produces 2.87 times more hydrogen than using only TiO₂. This noticeable enhancement is due to efficient charge separation in the presence of CFs with high visible light absorption. Comparatively, the CFs/TiO₂—S nanotexture produces H₂ at 2268.4 $\mu\text{mol g}^{-1}\text{h}^{-1}$, which is 6-12 fold greater than that for CFs/TiO₂ and pristine TiO₂ samples. Using the sol-gel approach, TiO₂ is effectively attached over the entire surface of CFs, enabling good interface interaction and efficient charge carrier separation, resulting in significantly enhanced H₂ production. Among the various operating parameters, glycerol as a sacrificial reagent promisingly improves H₂ yield, whereas it has a lower photostability than methanol after five consecutive cycles. The highest AQY of 26.3% for H₂ production is obtained with 150 mg of catalyst loading under low-intensity light irradiation.

[0061] The Carbon fibres (CFs) are synthesized/obtained through thermal decomposition of CFRPs, wherein they are separated from the polymer layer. The process of thermal decomposition at low temperatures yields highly conductive carbon fibers, which can be tailored into various sizes and shapes by eliminating the polymer layer. The schematic illustration for the synthesis of CFs using CFRPs is demonstrated in FIG. 1. The CFRPs serving as the raw material, is subjected to pyrolysis, the method involving evenly cutting the CFRPs sheets into a smaller size rectangular shape and heating the specific amount of CFRPs (placed in a ceramic boat) in a tube furnace under controlled nitrogen atmosphere at 500° C. for 2 h to remove polymeric material. This is followed by grinding the pyrolyzed product into small pieces and then pulverizing into fine powder using a planetary ball mill. In the experimental set up, the planetary ball mill PM100 is used. The one-dimensional carbon fibers (1D CFs) are synthesized using CFRP waste sheets.

[0062] In another embodiment of the invention, the CFRP serving as the raw material is subjected to pyrolysis, the method comprising evenly cutting CFRP sheets into smaller sized rectangular shapes (2 to 6 cm length and 1 to 4 cm width): placed in a ceramic crucible: heating the CFRPs in a tube furnace under controlled nitrogen atmosphere for removing polymeric material; and grinding the pyrolyzed product and pulverizing the ground product into fine powder.

[0063] In accordance with the present invention, CFs are coupled with TiO₂ through self-hybridization, and the photoactivity for H₂ production is tested with various CFs-loading. The role of interface interaction and morphology is investigated by synthesizing nanocomposites through physical mixing (CFs/TiO₂-A) and sol-gel methods (CFs/TiO₂—S). The chemicals and materials used for experiments are titanium (IV) isopropoxide (TTIP, purity 97%, Merck), acetic acid (CH₃COOH, MW=60.05 g/mol, 99.8%), 2-propanol (MW=60.1 g/mol,

CH₂CH(OH)CH₃, Merck), (titanium-(IV) oxide, anatase powder, 99.8%, Sigma-Aldrich), methanol (Merck, 99.8%), and carbon fiber-reinforced plastic.

[0064] In an embodiment of the present invention, the CFs/TiO₂ anatase composite is synthesized using the anatase phase of TiO₂ (titanium (IV) oxide, anatase powder). A direct physical mixing method is used to fabricate the CF-modified TiO₂ catalyst. The physical mixing method for CFs/TiO₂ composites is illustrated in FIG. 2. By using a single phase of mixing, different CF loadings onto TiO₂ are synthesized. Typically, methanol and TiO₂ are mixed for 2 h at room temperature. Subsequently, CFs with specific amounts are added and mixed for an additional 2 h to achieve good dispersion and interface interaction. The samples are subsequently dried (in an oven) for 24 h at 100° C. and are named as CFs/TiO₂ Anatase (CFs/TiO₂-A). Using the same procedure, varying CF loadings (1, 2, 3, 4, and 5 wt %) are added to TiO₂ suspension and given the label xCF/TiO₂-A, where x is the loading (1, 2, 3, 4, and 5 wt %). Carbon fiber loading and titanium dioxide anatase are denoted here by the acronyms xCF and TiO₂-A, respectively.

[0065] In another embodiment of the present invention, the CFs loaded over TiO₂ to get a good interface interaction are synthesized using the facile sol-gel approach. The scheme for the synthesis of various CF-based composites by sol-gel method is presented in FIG. 3. The precursor used for the synthesis of TiO₂ is titanium (IV) isopropoxide (TTIP). Typically, 10 mL of TTIP dispersed in 2-propanol is hydrolyzed using acetic acid (1 M) to get titanium sol. To get well-dispersed carbon fibers, CFs are added to the above titanium sol while it is being stirred, and this procedure is maintained for an additional 12 h to produce a thick titanium sol. Finally, the slurry is oven-dried at 100° C. for 24 h and heated at 500° C. for 2 h to get CFs/TiO₂ Sol-Gel (CFs/TiO₂-S) composite. The same process is used to synthesize TiO₂, but without the inclusion of CFs.

[0066] Another aspect of the proposed invention includes testing the synthesized CFs/TiO₂ composites with various CFs-loading for photoactivity for H₂ production. The performance of CFs/TiO₂ composites synthesized through physical mixing is compared against composites synthesized using sol-gel. Comparatively, CFs/TiO₂-S produced using the sol-gel method is more efficient due to good interface interaction and efficient charge carrier separation. Core-shell structure of nanoparticles with carbon fibres is achieved using the sol-gel method. In an embodiment of the invention, the performance of the CFs/TiO₂-S nanotexture is tested under various parameters and sacrificial reagents to maximize the hydrogen yield. Using glycerol as a sacrificial reagent with an optimized 3% CF loading results in the highest H₂ yield at lower catalyst loading levels. The photostability of the composite is evaluated over five consecutive cycles using glycerol and methanol as sacrificial reagents. With these findings, the nanocomposite can be optimized to improve its photocatalytic performance with the use of solid waste materials. The spent catalyst is further characterized to propose a mechanism for photocatalytic H₂ evolution over CFs/TiO₂ composite photocatalyst based on experimental and characterization outcomes.

[0067] In accordance with the present invention, the catalyst is characterized by using several analytical methods. The various analytical methods, experimental set up including the instruments for enabling the analytical characterization are described. In an embodiment of the present invention, the purity and crystal phase structures are examined using X-ray powder diffraction (XRD). To establish the interaction between the composite materials, Raman examination is carried out by using a scientific spectrophotometer (laser at 532 nm). Using scanning electron microscopy (SEM, JEOL), the structure and morphology are obtained. Transmission electron microscopy (TEM) is used to further explore the structure and morphology. The elemental state and composition are further assessed using XPS measurements (Axis Ultra DLD Shimadzu). A UV-visible diffuse reflectance spectrometer (DRS) is used to analyze the materials' optical response (V-750 UV-Visible Spectrophotometer). Spheres are used to load powder samples before the analysis.

The PL (photoluminescence) analysis is carried out using confocal micro-PL spectroscopy (Mode FEX-u) with a laser of wavelength 532 nm. The BET surface area is estimated using N₂ adsorption-desorption isotherms. Fourier-transform infrared spectroscopy (FTIR-4700, JASCO) is also used to understand the interaction of bonding and surface groups.

[0068] The present invention discusses the results of the analytical methods employed on catalytic sample. The XRD patterns of CFs, TiO₂ and CF-loaded TiO₂ samples are presented in FIG. 4. The peak at 2θ of 25.15° is ascribed to the crystalline phase of amorphous CFs with diffraction crystal plane (002), corresponding to the hexagonal primitive structure of carbon (JCPDS card no. 89-8487). This shows the polymer layer is successfully removed from the carbon fiber-reinforced polymers (CFRPs). The XRD patterns of TiO₂-A show peaks at 2θ of 25.39°, 37.87°, 48.05°, 53.96°, 55.21°, and 62.76° indexed to (101), (004), (200), (105), (211) and (204) facets, respectively (JCPDS card no. 21-1272) and confirms anatase TiO₂. In the case of TiO₂-S produced by the sol-gel method, similar peaks are obtained but with their lower crystallinity. An additional peak appears at 2θ of 27.55°, which is due to the formation of the rutile phase in TiO₂. In the case of CF/TiO₂-A, all the diffraction peaks of TiO₂-A appear; however, a peak belonging to CF does not appear due to their identical 2θ peaks. Similarly, all the peaks of TiO₂-S are present in the CF/TiO₂-S composite, and no additional peaks appear. These findings confirm successful fabrication of CF-loaded TiO₂ composites using both the physical mixing and sol-gel methods.

[0069] FIG. 5A shows Raman spectroscopy of pure CF, in which two high-intensity peaks are obtained at 1359.5 and 1599.1 cm⁻¹, depicted as the D and G bands of CF. The Raman spectra of TiO₂-A, TiO₂-S, and CF-loaded TiO₂ samples are presented in FIG. 5B. The Raman peaks for the TiO₂-A appear at 141.1, 196.4, 397.1, 517, and 639.2 cm⁻¹, which corresponds to pure anatase phase of TiO₂. For the case of TiO₂-S, the Raman spectroscopy peaks are obtained at 142.8, 196.8, 396.2, 517, and 640.9 cm⁻¹. When CF is added to TiO₂-A, the main peak appears at 142.3 cm⁻¹, which confirms a good interaction between TiO₂-A and CFs samples, confirming their successful fabrication. When CFs are loaded to TiO₂-S, similar patterns are obtained, which confirms successful fabrication of CF-based TiO₂ composites without any change in phase structure and without the presence of any other impurity.

[0070] The FTIR analysis of CF, TiO₂, and CF/TiO₂ samples is presented in FIG. 6. In the case of pure CF, no obvious peaks are obtained. The obvious vibration peaks at 2351 and 3739 cm⁻¹ is ascribed to C—H and hydroxyl bands, respectively. For the case of TiO₂-S. Ti—O—Ti vibrational frequencies appear at 517 and 703 cm⁻¹ and hydroxyl peaks appear at 1632 and 3439 cm⁻¹. For the case of TiO₂-A, the band at 726 cm⁻¹ belongs to the Ti—O—Ti vibrational frequency. In the case of CF-loaded TiO₂ composites, similar patterns are obtained.

[0071] The morphology of CF, TiO₂, and CF/TiO₂ samples is obtained through scanning electron microscopy (SEM). The TiO₂ anatase shows uniform size TiO₂ particles as shown in FIG. 7A. However, when the sol-gel method is used, agglomerated TiO₂ particles are obtained, as evidenced from FIG. 7B. All the particles are agglomerated together, whereas they have uniform size and shape. FIG. 7C-7D show the morphology of pure CFs. On analyzing the morphology of pure CFs, it is observed that large size carbon fibers are obtained, whereas the presence of other material is also identified, which would be due to breaking of large size fibers during grinding process of carbon fiber sheets. More importantly, fibers are covered with the amorphous materials as shown in FIG. 7D. The morphology of CF/TiO₂-A is presented in FIG. 7E-7F. It is noticed that CFs are well mixed with the TiO₂ particles, as shown in FIG. 7E. However, the high-resolution image in FIG. 7F shows that TiO₂-A is not spread uniformly over the entire surface of the CFs. This is possibly due to using a physical mixing approach, which is not successful to attach both the materials uniformly. Furthermore, the morphology of the

CFs/TiO₂-S composite, synthesized using the sol-gel method, as shown in FIG. 7G, presents that all of the TiO₂ particles are uniformly attached to the CF surface, with good interface interaction. The high-resolution image in FIG. 7H further confirms that the CF surface is entirely covered by the TiO₂ particles. Overall, the findings confirm successful fabrication of the CF/TiO₂-S composite, which is expected to promote interface charge carriers.

[0072] The morphology of the CFs/TiO₂-S is further investigated using transmission electron microscopy (TEM). The morphology of TiO₂ shows that all the TiO₂ particles are spherical in shape and well-distributed, confirming no agglomeration formation. Furthermore, uniformly sized TiO₂ NPs are attached to carbon fibers synthesized through sol-gel method. Evidently, carbon fibers are entirely covered by TiO₂ NPs, confirming good interface interaction between both materials. This is beneficial to promote charge separation during the photocatalysis process.

[0073] XPS examination is done to determine the elemental state of the CFs/TiO₂-S composite. FIG. 8A shows the results of Ti 2p high-resolution spectra with two obvious peaks positioned at 458.1 eV (Ti 2p_{3/2}) and 463.9 eV (Ti 2p_{1/2}), confirming the existence of titanium as Ti⁴⁺ or TiO₂. The XPS spectra of C 1s in FIG. 8B shows three peaks with binding energies 284.6, 286.1, and 288.2 eV, which are associated with C—C, C—O, and C=O bonds, respectively. This reveals that the CFs/TiO₂-S composite is successfully synthesized using a single-step sol-gel process. The survey spectrum in FIG. 8C is used to calculate the valence band position of TiO₂ which gives an estimated value of 2.60 eV (TiO₂ VB).

[0074] FIG. 9 shows UV-vis diffuse reflectance absorbance spectra of CF, TiO₂-A, TiO₂-S, CF/TiO₂-A and CF/TiO₂-S composite samples. The pure CFs has high light absorption in the visible region, which is evidently due to its high light absorbance capacity because of black color and higher conductivity. Comparatively, TiO₂-A shows higher light absorption in the visible region, whereas TiO₂-S has lower light absorption. This is possibly due to its large size particles, which have less light absorption efficiency compared to TiO₂ nanoparticles. The light absorption efficiency of TiO₂-S in the visible region is significantly increased when it is coupled with CFs. The CF/TiO₂-A composite, however, achieves the maximum light absorption in the visible region. As a result, the inclusion of CF in composites clearly increases their absorption in the visible light region, which is advantageous for increasing the activity of the photocatalyst. The results indicate that the presence of CFs in TiO₂ further enhances the light absorption efficiency, which is expected to improve the photocatalytic activity of the composite.

[0075] The band gap energy of the materials is calculated using Equation (1). The wavelengths of 378.8 and 383.3 nm are obtained for the TiO₂-S and 3CFs/TiO₂-S samples, giving band gap energies of 3.20 and 3.24 eV for the TiO₂-S and 3CFs/TiO₂-S samples, respectively. Previously, it was reported that CF did not display any absorption peak; however, when it is coupled with TiO₂, band gap energy gets reduced from 3.1 and 2.76 eV. However, in the current invention, band gap energy gets increased, which is due to increasing interface interaction through the use of the sol-gel method. The conduction band position is calculated using band gap energy and VB position of TiO₂ according to Equation (2).

$$[00001] E_{bg} = \frac{1240}{\lambda} \quad (1) \quad E_{CB} = E_{bg} - E_{VB} \quad (2)$$

[0076] Using Equation (2) with E_{VB} of 2.60 eV and E_{bg} of 3.20 eV, the E_{CB} of -0.60 eV is obtained. Overall, the analysis of absorbance spectra suggests that the composites have the potential to be effective photocatalysts, particularly for applications that require visible light activation. The E_{CB} value is promising to enhance reduction reaction for the production of hydrogen.

[0077] Photoluminescence (PL) studies are used to look at the effectiveness of electron-hole pair separation. In general, a high photoluminescence intensity is associated with low photocatalytic

activity and high electron-hole pair recombination efficiency. FIG. 10 shows PL analysis of CF, TiO₂-A, TiO₂-S, CF/TiO₂-A, and CF/TiO₂-S composite samples. In the case of pure CF, the lowest PL intensity is obtained, which is probably due to conductive characteristics and because of the blackish color. Using pristine TiO₂-S, the highest PL intensity is observed, whereas TiO₂-A shows lower PL intensity. This is most likely caused by large-sized TiO₂ particles, which have a greater rate of charge recombination than smaller TiO₂ particles. The PL intensity clearly decreases in the case of CF/TiO₂-A, which is evidently caused by the presence of CF, which inhibits the rate of charge recombination. Furthermore, a sol-gel process is used to synthesize the CF/TiO₂-S composite, which results in the lowest PL intensity yet recorded. This may have been brought about by effective charge carrier separation inside the CF/TiO₂-S composite because of good interface contact between the two materials. As a result, the PL peak intensity has been significantly decreased, which suggests that CF is able to prevent photoelectrons and holes from recombining. In many reports, decreased PL intensity of a semiconductor has been reported when coupled with other materials as cocatalyst. The PL spectrum of TiO₂/CF has been recently reported to be much lower than that of pristine TiO₂, indicating effective separation of electron-hole pairs. Similar to this, less PL intensity is seen when g-C₃N₄ and Ti₃C₂ are connected because the charge carriers separated more quickly. Another study found that the quicker charge separation caused the g-C₃N₄/Ti₃C₂ composite to have a lower PL intensity. These results all support the idea that CF can function as a cocatalyst and that photogenerated electrons can be effectively transported from TiO₂ to CF, which is beneficial to stimulate the hydrogen production efficiency.

[0078] To further study the BET specific surface area and pore volume, nitrogen adsorption-desorption isotherm analysis is performed. FIG. 11 shows the isotherms of TiO₂ and 3% CFs/TiO₂-S samples, which belong to type IV with H₃ hysteresis loops, confirming the presence of a mesoporous structure. The BET surface area and BJH pore volume of all the samples are summarized in Table 1. Pure TiO₂ has a BET surface area of 43 m²/g, whereas loading CFs to TiO₂ has reduced the specific surface area to 16 m²/g. This decrease in BET surface area is probably due to lower surface area of CFs because the polymeric layer is not completely removed, thus acting only as a support for the TiO₂. Comparatively, the pore volume of pure TiO₂ is 0.060 cm³/g, which is increased to 0.10 cm³/g when 3% CF is added to TiO₂. This reveals that CFs do not support increased BET surface area but they are beneficial to increased total pore volume. Previously, commercial CFs/TiO₂ composite was tested and reported a much higher surface area of the composite (716.8 m²/g) compared to pure TiO₂ (54 m²/g). In the present invention, an increase in photocatalytic activity would not be due to increasing BET surface area, but it would be due to proficient charge carrier separation as evidenced by PL analysis.

TABLE-US-00001 TABLE 1 Summary of BET Surface Area and Pore Volume of TiO₂-S and the 3% CFs/TiO₂-S Composite Catalyst

	BET surface area (m ² /g)	BJH pore volume (cm ³ /g)
TiO ₂	43	0.060
3% CFs/TiO ₂	16	0.122

[0079] In an embodiment of the present invention, photoactivity test is performed on the synthesized composites. The performance of pure TiO₂ and carbon fiber-based nanocomposite is examined using a slurry phase Pyrex glass photoreactor system with a double cylinder having a total volume of 120 mL. A 35 W xenon lamp served as the visible light source with a light intensity of 20 mW/cm². A 100 mg sample of the photocatalyst is disseminated in a 5 vol % aqueous solution of methanol and homogenized by magnetic stirring. Utilizing a vacuum flow system and constant nitrogen flow, the reactor and piping system are cleaned, and the efficiency is determined based on the amount of hydrogen produced. Several loadings of catalysts (50-200 mg) are used in a 100 mL solution of 95 mL deionized water and 5 mL methanol, and the mixture is constantly agitated on a magnetic stirrer. In different embodiments, in order to investigate the effect of sacrificial reagents, various sacrificial reagents such as methanol, ethanol, butanol, and glycerol are

used with the concentration of 5 vol %.

[0080] The online micro-GC (fusion) is integrated with the reactor for the continuous analysis of the amount of H₂ produced. The GC instrument is installed with two TCDs connected with argon and helium carrier gases, and products are injected into the GC instrument after a 15 min interval. Following the evaluation of photocatalytic hydrogen production activities, a stability test is conducted for a total of 5 cycles of 2 h each. For 1 h after and before each experiment, N₂ gas is flown through the reactor. The used catalyst after five cycles is dried at 100° C. for 24 h before further characterization analysis. The results of photocatalytic test for hydrogen production are discussed in the following sections.

[0081] In a slurry photoreactor with 5% methanol as the sacrificial reagents and 100 mg of catalyst loading, the performance of TiO₂-A and CF/TiO₂-A with varied CF loadings (1 to 5 wt %) is tested. FIG. 12A shows photocatalytic H₂ production over different CF-loaded TiO₂-A samples. In the presence of pure CFs, production of hydrogen is not detected due to its conductive characteristics. Using pure TiO₂-A, the H₂ yield of 59.57 ppm is produced after 2 h of irradiation time. When TiO₂-A is coupled with a very small amount of CF (1 wt %), the hydrogen yield gets increased to 77.65 ppm. The highest H₂ production of 170.82 ppm is obtained with a 3% CF/TiO₂-A composite. This amount of H₂ production is 2.87-fold higher than using pristine TiO₂-A. Due to the CFs' conductive properties, which effectively separate photoinduced electrons and holes over the TiO₂ surface, there is a notable increase in hydrogen productivity. However, any loading of CF above 3 wt % has adverse effect on the hydrogen evolution, whereas the lowest H₂ yield is obtained with 5 wt % CF loading with TiO₂-A. These trends can be explained based on charge production and separation and their recombination over the CF/TiO₂-A composite. At lower CF loading, more electrons are trapped by CF due to its higher conductivity and good interaction with the TiO₂-A surface. However, when CF-loading is increased above the optimized value, more CFs interact with each other instead of TiO₂, resulting in recombination of charges transferred toward the CF surface. Another important factor can be the lower surface area of TiO₂-A available for producing photoinduced charge carriers. In both possibilities, lower photoinduced charge carriers will be available for oxidation and reduction reactions; thus, a lower H₂ yield is produced.

[0082] Further research into the performance of CF-loaded TiO₂ is conducted by utilizing several synthesis techniques, including the physical mixing of CFs and TiO₂-A to prepare 3CFs/TiO₂-A and the sol-gel approach to produce nanotextures of CFs/TiO₂—S. FIG. 12B displays the comparison of these photocatalytic systems with 5 vol % methanol and 100 mg of catalyst loading. Using 3CFs/TiO₂-A, the hydrogen production gets decreased after 60 min of irradiation time. However, in the case of 3CFs/TiO₂—S, produced with the sol-gel method, a continuous yield of H₂ is obtained over the entire irradiation time. Initially, for the first 40 min, the lower yield of H₂ is obtained, which gets significantly increased after 60 min. The lower amount of hydrogen production at the beginning could be possible due to the existence of an activation process in the CFs/TiO₂ composite, which can strengthen the interaction between TiO₂ and CFs under continuous stirring. Another possible reason could be due to using continuous process with online injection of samples to the GC system after a 15 min interval. In this case, there is no accumulation of hydrogen in the reactor and pipelines which are connected to the GC instrument.

[0083] Using optimized 3% CFs/TiO₂—S, H₂ production reaches a maximum amount of 907 ppm after 2 h of irradiation time, which is 6-fold more than is produced using the same CF loading but with TiO₂-A through a physical mixing approach (3CFs/TiO₂-A). According to SEM analysis, there is no good interface interaction between CF and TiO₂ using the physical mixing method; thus, CFs are not well connected with TiO₂, enabling lower transfer of electrons, resulting in trivial photocatalytic activity for H₂ production. For the sol-gel method, CFs are entirely covered with TiO₂ particles, resulting in the efficient transfer of

electrons to maximize the yield of H₂.

[0084] The schematic illustration for photocatalytic H₂ evolution over 3CFs/TiO₂-A and 3CFs/TiO₂-S is presented in FIG. 12C and FIG. 12D. The interaction between the cocatalyst and the semiconductor is important to effectively transfer electrons during the photocatalytic process. In the current case, CFs work as the sink, and electrons are transferred from TiO₂ to CFs due to their higher conductive characteristics. However, due to lower interface interaction in CFs/TiO₂-A, the process of charge transfer is not efficient, as illustrated in FIG. 12C. On the other hand, using CFs/TiO₂-S, TiO₂ particles are entirely and uniformly attached to CFs to give good interface interaction as shown in FIG. 12D. Due to the good interface interaction between CFs and TiO₂-S, efficient charge transfer would be possible, which enables 6-fold increments in H₂ yield compared to using CFs/TiO₂-A composite. CF/TiO₂ is tested for photocatalytic water splitting and results in an enhanced photoactivity of TiO₂ with CF loading due to increased BET surface area, higher visible light absorption, and efficient charge carrier separation. In the current invention, enhanced photocatalytic H₂ production is obviously due to higher light absorption and proficient charge carrier separation.

[0085] Sacrificial reagents are known to be crucial in water splitting applications; hence, the performance of the improved 3CFs/TiO₂-S is further examined utilizing a variety of sacrificial reagents. In different embodiments of the invention, the sacrificial reagents employed comprises methanol, ethanol, butanol, and glycerol with 5 vol % concentration and 100 mg catalyst loading, and results are presented in FIG. 13. Using pure water, production of hydrogen is not significant, due to more charge recombination and the unavailability of protons to promote hydrogen evolution. However, when sacrificial reagents of 5 vol % are used with water, a significant amount of H₂ is produced. This is because sacrificial reagents have a major impact on the photocatalyst activity and serve as hole scavengers. Obviously, using different types of sacrificial reagents has different effects: the highest H₂ yield is obtained using glycerol, and the lowest amount of H₂ is produced with butanol. With 3CFs/TiO₂-S, glycerol produces the largest amount of H₂ (1650 ppm), which is 1.8, 2.3, and 17.2 times more than those of methanol, ethanol, and butanol, respectively. Generally speaking, standard oxidation potential and the quantity of OH bonds all have a significant impact on H₂ generation. Alcohols enhance polarity and thus have a favorable impact on H₂ evolution. In the order of butanol (C₄H₁₀O), ethanol (C₂H₆O), methanol (CH₄O), and glycerol (C₃H₈O₃), H₂ generation is seen to rise. Furthermore, α -hydrogen atoms present in alcohols enable more H₂ production. The molecular structures of glycerol and methanol consist of more carbon atoms attached to α -hydrogen atoms; thus, they are more favorable to enhance H₂ production. Previously, when investigated various sacrificial reagents and found hydrogen production in the order TEOA>methanol>glycerol. The discrepancy of the results in the current invention is due to using TiO₂, which has more potential to absorb glycerol compared to g-C₃N₄ which has more potential to absorb TEOA and methanol than glycerol.

[0086] In photocatalysis, catalyst type, catalyst loading, type of reactor, and light intensity are the influential parameters responsible for maximizing the amount of H₂ evolution. Therefore, glycerol is used as the sacrificial agent while the performance of improved 3CFs/TiO₂-S is further examined, and the findings are described in FIG. 14. With 50 mg of the catalyst, 1260 ppm of H₂ production is obtained, whereas H₂ yield gets further increased when catalyst loading is increased to 100 mg (1651 ppm), which is 1.31 times more than using 50 mg of catalyst. This obvious enhancement in photocatalytic H₂ evolution is due to more surface being available for the oxidation and reduction reactions over the 3CFs/TiO₂-S composite catalyst. However, when catalyst loading is increased to 150 mg, the H₂ production increases to 2790 ppm, which is 1.70 and 2.2 fold higher than using 100 and 50 mg loading amounts. However, any further increase in catalyst loading (200 mg) does not have much effect on enhancing H₂.

yield. All these findings confirm that the catalyst loading of 150 mg is the optimized amount for glycerol sacrificial reagents, at which the highest amount of H₂ is evolved.

[0087] Based on the surface area available for the oxidation and reduction process, all of the data can be explained. There is only a small quantity of catalyst (50 mg) particles available for the reaction to occur under the source of light, which leads to a decrease in the generation of electrons and holes-both of which are crucial for redox processes. There is a decent dispersion of nanoparticles to interact with light irradiation when the catalyst loading is raised to 100 and 150 mg, allowing for additional charges to be generated for the evolution of hydrogen. However, an excessive catalyst loading might result in a light penetration resistance; as a result, when catalyst loading is raised to 200 mg, there is no appreciable impact on the generation of H₂, and similar patterns were previously observed. All these data support the notion that catalyst loading is another crucial factor that should be tuned for maximum photocatalytic performance under the same operating circumstances.

[0088] In the photocatalysis process, two different variables, namely, catalyst loading and light intensity, are used; thus, comparing performance of different photocatalytic system requires a more efficient approach. Calculating quantum efficiency is a more reliable method because it is based on the number of moles of products produced and the moles of photons consumed during the photocatalytic process. Thus, the relationship between the rate of production of H₂ and rate of photon flux ingested over the course of that particular unit of time is known as apparent quantum yield (QY), which can be calculated using Equation (3).

$$[00002] \text{AQY of } H_2 (\%) = \frac{2 \times \text{moles of } H_2 (\text{mole / time})}{(\text{incident area} \times \text{photon flux}) (\text{mole / time})} \times 100 \quad (3)$$

where the photon flux is calculated using the following parameters: wavelength of 420 nm, light intensity of 20 mW cm⁻², incident area of 28 cm², moles of H₂ would be in μmol/s, and the number of electrons utilized to produce H₂ is 2.

[0089] Table 2 provides a summary of the hydrogen production rate and apparent quantum yield for TiO₂, 3CFs/TiO₂-A, and 3CFs/TiO₂-S with different sacrificial reagents and catalyst loading. Hydrogen production rate is calculated after 2 h of irradiation time. Using pure TiO₂, an H₂ rate of 224.1 μmol g⁻¹ h⁻¹ is produced, which gets increased to 370.4 μmol g⁻¹ h⁻¹ with 3% CFs-loading (3% CFs/TiO₂-A). Comparatively, when catalysts are produced through sol-gel method, the highest H₂ production rate of 2268.4 μmol g⁻¹ h⁻¹ with QY of 6.629% over 3% CFs/TiO₂-S is obtained, which is 6.12 and 10.12 times more than is produced using 3% CFs/TiO₂-A and pure TiO₂ samples, respectively. This obvious improvement is due to the good interface interaction of CFs and TiO₂ NPs with proficient charge carrier separation and higher visible light absorption. Using 5% glycerol sacrificial reagent and 100 mg catalyst loading, an H₂ production rate of 4126.1 μmol g⁻¹ h⁻¹ and AQY of 12.058% are achieved, which are 1.82, 2.31, and 17.24 fold higher than those using methanol, ethanol, and butanol, respectively. More interestingly, when catalyst loading is increased, results are different, whereas the highest H₂ production is obtained with optimized catalyst loading of 150 mg for both glycerol and methanol sacrificial reagents. Comparatively, the highest hydrogen production of 6000.3 μmol g⁻¹ h⁻¹ and QY of 26.302% is achieved with 5% methanol and 150 mg catalyst loading. This amount of hydrogen production is 1.83 fold higher than using glycerol as the sacrificial reagents. These findings reveal that at lower catalyst loading, glycerol is more efficient for H₂ production, whereas, with increasing catalyst loading, methanol is the proficient sacrificial reagent due to adsorption competitions and the rate of intermediate products over the catalyst active sites. The highest AQY of 26.3% for H₂ production is obtained with 150 mg of catalyst loading under low-intensity light irradiation, whereas it has a lower photostability than methanol after five consecutive cycles.

TABLE-US-00002 TABLE 2 H₂ Production Rate and Quantum Yield over TiO₂,

CF/TiO₂.sub.2-A, and CF/TiO₂.sub.2-S Composites with Different Sacrificial Reagents and Catalyst Loading Sacrificial reagent Catalyst loading H₂.sub.2 production rate Quantum yield Catalyst (vol %) (mg) (μmole g^{sup.}.-1 h^{sup.}.-1)* (QY, %) TiO₂.sub.2 5% Methanol 100 224.1 0.007 3CFs/TiO₂.sub.2-A 5% Methanol 100 370.4 1.083 3CFs/TiO₂.sub.2-S 5% Methanol 100 2268.4 6.629 3CFs/TiO₂.sub.2-S 5% Glycerol 100 4126.1 12.058 3CFs/TiO₂.sub.2-S 5% Ethanol 100 1783 0.052 3CFs/TiO₂.sub.2-S 5% Butanol 100 239.4 0.699 3CFs/TiO₂.sub.2-S 5% Glycerol 150 4650.6 0.204 3CFs/TiO₂.sub.2-S 5% Glycerol 200 3287.1 19.211 3CFs/TiO₂.sub.2-S 5% Methanol 150 6000.3 26.302 *Hydrogen production rate calculated after 2 h of irradiation time.

[0090] In another embodiment of the present invention, the synthesized CF/TiO₂.sub.2 composites is examined for stability analysis. Examining the stability and reusability of the photocatalysts is an additional crucial issue that should be looked at considering their practical use and applications. Glycerol and methanol are used as the sacrificial reagents, and their 5 vol. percent concentrations are used to test the stability of 3CFs/TiO₂.sub.2—S composite with 150 mg catalyst loading. After every completed cycle, the reactor is allowed to cool, and the hydrogen is removed from the pipeline and inside the reactor with nitrogen gas before starting the next experiment. FIG. 15A shows the stability of the photocatalyst with glycerol as the sacrificial reagent for the consecutive five cycles. Using the first cycle, the highest H₂.sub.2 yield is obtained, and its production is continuous over the entire irradiation time. Unexpectedly, a sudden drop in the H₂.sub.2 yield is observed in the second cycle, whereas a stable H₂.sub.2 yield is produced until the fifth cycle. These results can be explained based on the characterization results. Initially, there is a good interface interaction of 3CFs/TiO₂.sub.2—S, whereas when it is dispersed inside the glycerol through stirring, the CFs possibly are separated from the TiO₂.sub.2, resulting in good interface interaction, which results in a decline in H₂.sub.2 yield after the first cyclic run. However, a stable and continuous H₂.sub.2 yield for the remaining cycles is possibly due to no more segregation of CFs from the TiO₂.sub.2 surface.

[0091] The stability of 3CFs/TiO₂.sub.2—S for photocatalytic H₂.sub.2 evolution with 5 vol % methanol with consecutive 5 cycles is presented in FIG. 15B. In the first cyclic run, the highest amount of H₂.sub.2 is produced over the entire irradiation time, and similar trends are obtained in the second cyclic run. However, after the second cyclic run, a sudden drop in H₂.sub.2 yield is observed, whereas a stable performance of the photocatalyst is obtained in remaining cycles until the fifth cycle. By comparing these results with the stability of photocatalyst using glycerol, identical trends are observed. More interestingly, using methanol, a sudden drop in H₂.sub.2 yield is observed in the second cycle, whereas for the glycerol after the first cycle, a sudden drop in H₂.sub.2 yield is observed. These results can be explained based on the viscosity and attachment of catalyst over the catalyst surface. Due to the high viscosity of glycerol, there is more resistance for the catalyst particles to disperse inside the slurry, resulting in their early separation of CFs from the TiO₂.sub.2 in comparison to using methanol. However, further investigations are required to further explore the sudden drop in photostability using various sacrificial reagents.

[0092] The decrease in photostability is further investigated using XRD, FTIR, and SEM analysis of the spent 3CFs/TiO₂.sub.2—S composite samples. XRD analysis is performed to investigate the change in crystallinity and composition of the photocatalyst after five consecutive cycles, and the results are presented in FIG. 16. The same peak locations between fresh and used samples demonstrate that the peak position has not changed. New peaks are not observed in the spent composite sample, further supporting the stability and lack of structural change. Previously, XRD analysis of TiO₂.sub.2/CF was conducted after 10 cycles, which kept the same structure as the original materials. Additionally, the photocatalyst TiO₂.sub.2 contents remained the same from the first to the tenth cycling trials, indicating that the CF/TiO₂.sub.2 material was sufficiently stable to produce H₂.sub.2.

[0093] FIG. 17 shows FTIR analysis of the spent CFs/TiO₂.sub.2—S composite after stability analysis of glycerol and methanol. Both the samples show similar IR spectra; however, some

changes are observed in the vibrational modes. For example, more hydroxyl bonds are observed in the composite catalyst obtained after glycerol stability analysis, whereas using methanol, no obvious peaks related to hydroxyl group attachments are observed. This may be possibly due to more moisture content in the spent catalyst obtained after glycerol stability analysis, which is difficult to evaporate during the drying process at 100° C. In general, both the samples show similar IR peaks, confirming there are no significant structural changes in both the samples.

[0094] The morphology of the spent photocatalyst obtained after 5 cycles using glycerol as the reducing agent is further investigated, and the EDX spectrum results are shown in FIG. 18. There is a good dispersion of CF with TiO₂-S in the composite, which still enables the production of a higher amount of H₂. TiO₂ particles are separated from the CF surface, which are possibly responsible for the decrease in hydrogen yield after the first cycle. This is further confirmed using EDS mapping analysis. It can be observed that the dispersion of carbon with Ti and O is not uniform, which is possibly due to segregated CFs from the TiO₂ due to continuous stirring and agitation in the glycerol-water mixture. FIG. 18 shows the EDX spectrum, showing the presence of all of the elements in the CF/TiO₂ composite. The higher composition of C is possibly due to selecting small diffraction area for EDX analysis, where a greater amount of CF is present.

[0095] In another embodiment of the present invention, a mechanism is proposed for photocatalytic H₂ evolution over a 3CFs/TiO₂-S composite photocatalyst.

[0096] In photocatalysis, information about the charge production and separation is important to understand more about the process of efficiency enhancement. According to experimental results, CF/TiO₂-A has lower photoactivity compared to CF/TiO₂-S composite synthesized using the sol-gel approach. In the case of CF attached to TiO₂ using physical method, there is only point contact between CF and TiO₂; thus, only a lower transfer of electrons is possible toward the CF. However, in the case of 3% CFs/TiO₂-S, TiO₂ particles are dispersed everywhere over the entire CF surface, thus providing multichannel electron transfer, enabling efficient charge carrier separation. In previous work, similar observations were reported during investigating the effect of morphology and particle size of TiO₂ with other semi-conductors.

[0097] The schematic illustration of hydrogen production over 3% CFs/TiO₂-S is further discussed in FIG. 19. Under light energy, electron (e⁻) and holes (h⁺) are generated over the TiO₂ surface, which can either recombine or can involve various oxidation and reduction reactions. Therefore, the available electrons at the CB of the semiconductor can affect the production of hydrogen, whereas H₂ yield can be increased by preventing the recombination of electrons and holes. According to BET analysis, loading of TiO₂ over CFs reduces the specific surface area; however, PL analysis reveals lower intensity in the composite compared to using pure TiO₂, which is due to efficient charge carrier separation. Thus, due to the good interface interaction of TiO₂ with CFs as cocatalyst, electrons would effectively be trapped by CF due to its conductive characteristics, resulting in efficient charge carrier separation. The holes retained at the VB of TiO₂ will be consumed for water and/or alcohol oxidation to produce protons (H⁺). The electrons retained at the CBs and trapped by CFs will be used for the reduction of H⁺ to produce H₂. The conduction band of TiO₂ (CB≈-0.60 eV) calculated previously is higher than the oxidation potential of H⁺/H₂ (-0.42 eV); thus, this process is favorable to produce hydrogen over the CFs/TiO₂ composite. Thus, due to the conductive characteristics of CFs, electrons would effectively be transferred from TiO₂ toward CFs, resulting in their efficient separation and utilization for reduction of protons to produce H₂.

[0098] The recombination of photogenerated electrons and holes can be further prevented by using sacrificial reagents, which use holes. The VB holes will oxidize alcohol (Alcohol+h⁺→oxidation products), resulting in effective utilization of holes and preventing charge recombination. Recently, it was reported that TiO₂/CF has higher

photocatalytic H₂ evolution rate due to increased BET surface area, higher visible light absorption, and lower charge recombination rate. Thus, to considerably improve the photocatalytic H₂ evolution, CFs/TiO₂ is found promising because of their special features such as higher visible light absorption and conductive characteristics to promote photocatalytic activity. [0099] In view of cost estimation, the carbon fibers (CFs) are very low-cost materials compared to expensive metals like gold (Au) and platinum (Pt) due to using carbon fiber-reinforced polymers (CFRPs). The CFRPs is a solid waste product (sewage sludge derived from the CFRP waste) which is produced by the industry in tonnes every day. On the other hand, TiO₂ is already a commercial catalyst, which is produced using low-cost precursors through different methods. Thus, using CFs/TiO₂ has more economic benefits compared to using expensive metals with TiO₂ to produce hydrogen.

[0100] In summary, the conversion of carbon fiber-reinforced plastics (CFRPs) into carbon fibers (CFs) through thermal decomposition shows great potential for enhancing visible light absorption with higher charge separation efficiency. The thermal decomposition method at low temperatures can derive highly conductive carbon fibres of different sizes and shapes by removing the polymer layer. The use of CF/TiO₂-A composite results in significantly higher hydrogen (H₂) production due to the higher charge separation and visible light absorption efficiency of CFs. The highest H₂ production of 170.82 ppm is achieved with the 3% CF/TiO₂-A composite, which is 22.87 fold higher than using pristine TiO₂-A. Further investigation into CF/TiO₂ performance using different synthesis methods, namely physical mixing and sol-gel approach, reveals that amount of H₂ yield over 3% CF/TiO₂-S is 6 fold more than is produced using the same CF loading but with TiO₂-A through a physical mixing approach (3% CF/TiO₂-A).

[0101] The optimized 3CFs/TiO₂ produces 2.87 times more hydrogen than using only pristine TiO₂. This noticeable enhancement is due to efficient charge separation in the presence of CFs with high visible light absorption. Comparatively, the CFs/TiO₂-S nanotexture produced H₂ at 2268.4 $\mu\text{mol g}^{-1} \text{h}^{-1}$, which is 6-12 fold greater than that for CFs/TiO₂ and pristine TiO₂ samples. Using the sol-gel approach, TiO₂ is effectively attached over the entire surface of CFs, enabling good interface interaction and efficient charge carrier separation, resulting in significantly enhanced H₂ production.

[0102] In another embodiment of the present invention, various operating parameters and sacrificial reagent are used to enhance H₂ production. Among the various operating parameters used, glycerol as a sacrificial reagent is effective in enhancing H₂ yield, whereas it has a lower photostability than methanol after five consecutive cycles. The highest AQY of 26.3% for H₂ production is obtained with 150 mg of catalyst loading under low-intensity light irradiation. The invention presents a novel method of recycling solid waste CFRPs to produce CF-based composites for enhancing performance in solar energy-related applications.

[0103] This significant enhancement in photoactivity is attributed to the good interface interaction between CF and TiO₂ achieved through the sol-gel approach, leading to enhanced photocatalytic activity for H₂ production. Glycerol is one of the most effective sacrificial reagents, yielding higher H₂ compared to methanol, ethanol, and butanol. Among the catalyst loading, the H₂ yield increases with more catalyst loading, until reaching an optimized loading amount of 150 mg. The stability of the 3% CFs/TiO₂-S composite is investigated using glycerol and methanol as sacrificial reagents, revealing similar trends where stability drops after the first cycle but there is continuous H₂ evolution for 5 cycles. This invention introduces a novel method for recycling CFRP and constructing CF-based TiO₂ composites, offering development of highly effective photocatalysts and solar energy applications. It has the potential to accelerate the development of renewable energy sources and reduce dependency on fossil fuels.

[0104] Solid waste such as aircraft waste, and car waste can be used to produce high-quality carbon fibres. Carbon fibre can be incorporated as a sensitizer in dye-sensitized solar cells to increase

efficiency and stability.

[0105] It will be apparent to those skilled in the art that various modifications and variations can be made in the present invention without departing from the spirit or scope of the inventions. Thus, it is intended that the present invention covers the modifications and variations of this invention provided they come within the scope of the appended claims and their equivalents. The disclosures and the description herein are intended to be illustrative and are not in any sense limiting the invention, defined in scope by the following claims. Many changes, modifications, variations and other uses and applications of the subject invention will become apparent to those skilled in the art after considering this specification and the accompanying drawings, which disclose the preferred embodiments thereof. All such changes, modifications, variations and other uses and applications, which do not depart from the spirit and scope of the invention, are deemed to be covered by the invention, which is to be limited only by the claims which follow.

[0106] Many changes, modifications, variations and other uses and applications of the subject invention will become apparent to those skilled in the art after considering this specification and the accompanying drawings, which disclose the preferred embodiments thereof. All such changes, modifications, variations and other uses and applications, which do not depart from the spirit and scope of the invention, are deemed to be covered by the invention, which is to be limited only by the claims, which follow.

Claims

1. A method of developing semiconductor photocatalysts by recycling Carbon Fiber-Reinforced Polymers (CFRP) waste, the method comprising the steps of: separating or removing a polymer layer of carbon-fibre waste via thermal decomposition; and coupling the resulting carbon-fibres as a cocatalyst with semiconductor materials; for photocatalytic water splitting results in producing/generating hydrogen (H.sub.2).
2. The method of claim 1, wherein carbon-fibre (CF) is synthesized by being separated from the polymer layer via thermal decomposition of CFRPs, at temperatures ranging between 400-700° C., yielding highly conductive carbon fibers (CFs).
3. The method of claim 2, wherein the CFs are tailored into various sizes and shapes by eliminating the polymer layer.
4. The method of claim 2, wherein the CFRP serving as the raw material, is subjected to pyrolysis, the method comprising the following steps: evenly cutting CFRP sheets into smaller sized rectangular shapes; heating the CFRPs in a tube furnace under a controlled nitrogen atmosphere for removing polymeric material; and grinding the pyrolyzed product and pulverizing the ground product into fine powder.
5. The method of claim 4, wherein the controlled nitrogen atmosphere comprises a temperature of 500° C. for 2 hours.
6. The method of claim 4, wherein the smaller sized rectangular shapes are 2-6 cm in length and 1-4 cm in width.
7. The method of claim 4, wherein pulverizing the ground product into fine powder is done using a planetary ball mill.
8. The method of claim 1, wherein coupling the resulting carbon-fibres as a cocatalyst with semiconductor materials is done via facile hydrothermal methods such as sol-gel (CFs/TiO.sub.2—S) and ultrasonic/physical mixing approaches (CFs/TiO.sub.2—A).
9. The method of claim 8, wherein using the sol-gel approach, TiO.sub.2 is attached over the entire surface of CFs, enabling interface interaction and charge carrier separation, resulting in enhanced H.sub.2 production.
10. The method of claim 8, wherein using the sol-gel approach results in achieving a core-shell structure of nanoparticles with carbon fibres.

11. The method of claim 1, wherein the CFRP waste-derived carbon fibers (CFs) is coupled with TiO.sub.2 to construct CFs/TiO.sub.2 nanotextures with efficient interface charge transfer for stimulating photocatalytic hydrogen production.

12. A carbon fibre (CF)/TiO.sub.2 composite, wherein the carbon fiber is derived from carbon fiber-reinforced polymer (CFRP) waste by separating a polymer layer from the CFRP waste via thermal decomposition under a controlled nitrogen atmosphere, wherein the TiO.sub.2 nanoparticles are uniformly and continuously attached to the surface of the carbon fibers to form a core-shell structure, and are spherical, well-distributed, and free of agglomeration, wherein the composite is synthesized by a sol-gel process in which titanium dioxide is deposited over the entire surface of the carbon fibers, wherein the resulting composite exhibits interface interaction and charge carrier separation between the carbon fibers and the TiO.sub.2 nanoparticles, wherein the composite is configured for use as a photocatalyst for photocatalytic hydrogen production via water splitting, and wherein the carbon fibers are tailored into multiple and varying sizes and shapes by eliminating the polymer layer during the thermal decomposition process.

13. The CF/TiO.sub.2 composite of claim 12, wherein the thermal decomposition of CFRPs occurs at a temperature between 400° C. and 700° C.

14. The CF/TiO.sub.2 composite of claim 12, wherein the CF/TiO.sub.2 composite is synthesized using the sol-gel method, wherein XPS spectra of C shows three peaks with binding energies 284.6, 286.1, and 288.2 eV, which are associated with C—C, C—O, and C=O bonds, respectively.

15. The CF/TiO.sub.2 composite of claim 12, wherein the CF/TiO.sub.2 composite is synthesized using physical mixing approaches.

16. The CF/TiO.sub.2 composite of claim 12, wherein morphology of the CFs/TiO.sub.2 is investigated using one or more of transmission electron microscopy (TEM), X-ray powder diffraction (XRD), Raman spectroscopy and scanning electron microscopy (SEM).

17. The CF/TiO.sub.2 composite of claim 12, wherein the CF/TiO.sub.2 composite promotes charge separation and visible light absorption during the photocatalysis process.

18. The CF/TiO.sub.2 composite of claim 12, wherein glycerol is used as a sacrificial reagent to improve hydrogen (H.sub.2) yield.

19. The CF/TiO.sub.2 composite of claim 18, wherein an apparent quantum yield (AQY) of 26.3% for H.sub.2 production is obtained with 150 mg of catalyst.

20. The CF/TiO.sub.2 composite of claim 12, wherein the CF is incorporated as a sensitizer in dye-sensitized solar cells to increase efficiency and stability.
